



INVENTORY MANAGEMENT SYSTEM: FOR SMALL BUSINESSES

Presented by Harkirat, Diya and Ananya



PROBLEM STATEMENT

Small businesses such as grocery stores, pharmacies, and retail shops often rely on manual methods or basic spreadsheets to track inventory. This leads to challenges such as:

- Inaccurate stock levels due to human error in manual entries.
 - Frequent stock outs or overstocking, causing either lost sales or wasted resources.
 - Time-consuming reporting, as business owners must prepare invoices and summaries manually.
- 



OUR SOLUTION

Reorder Alerts:

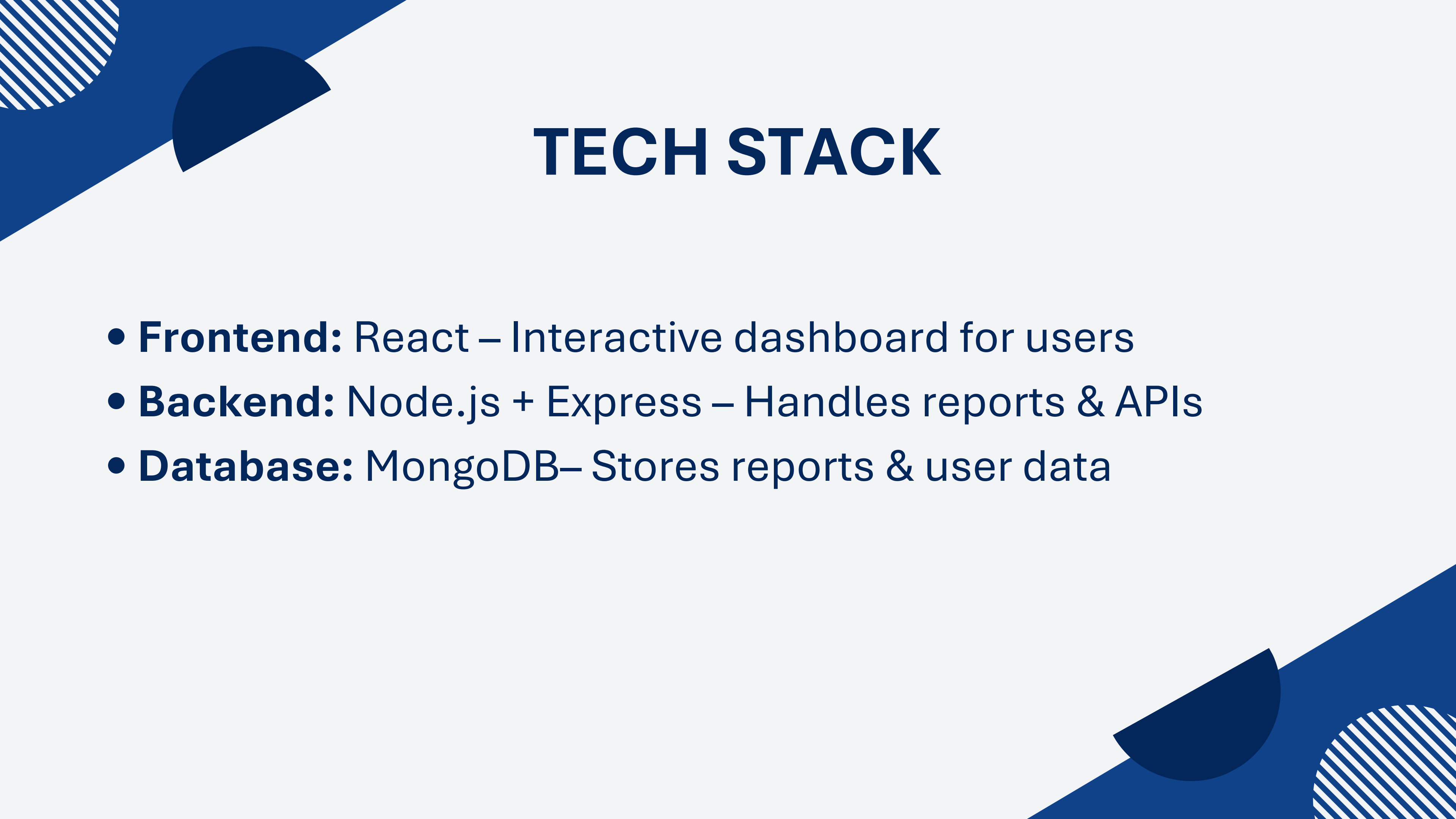
Set threshold levels for products. When stock falls below, the system sends automated alerts or generates purchase orders so businesses never run out of popular items

Reporting and Analytics:

Generate inventory reports and sales analytics for decision-making. Dashboard visualizations of stock status, top-selling products, and reorder forecasts help plan business needs.

Barcode Scanning Integration:

Use barcode scanners or mobile camera as barcode readers to speed up adding and updating inventory. This reduces manual entry errors and accelerates checkout and stock count.



TECH STACK

- **Frontend:** React – Interactive dashboard for users
- **Backend:** Node.js + Express – Handles reports & APIs
- **Database:** MongoDB– Stores reports & user data

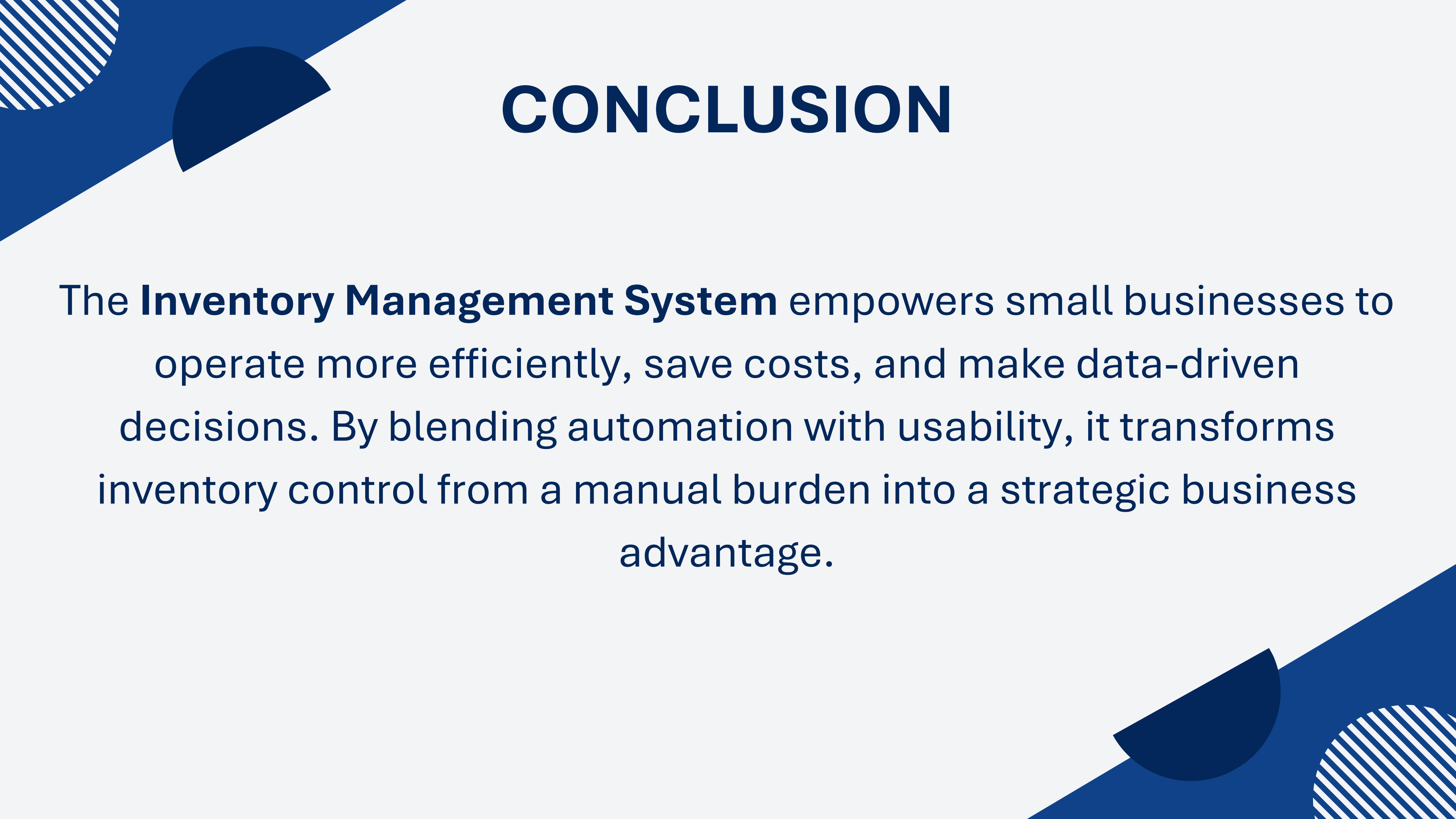
USP AND IMPACT

- **Unique Selling Proposition (USP):**

- Unlike generic inventory tools, this system combines real-time stock tracking, automated reorder alerts, barcode integration, and analytics in one lightweight, scalable platform designed specifically for small business needs.
- Its simplicity, affordability, and automation give owners enterprise-grade efficiency without the complexity of big ERP systems.

- **Impact:**

- Reduces losses from stockouts/overstocking.
- Saves time & effort by automating manual inventory and reporting tasks
- Improves decision-making with sales trends and demand insights.



CONCLUSION

The **Inventory Management System** empowers small businesses to operate more efficiently, save costs, and make data-driven decisions. By blending automation with usability, it transforms inventory control from a manual burden into a strategic business advantage.



QNA SESSION



**THANK
YOU**

SMART INVENTORY SYSTEM :

For Small Businesses

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NEED ANALYSIS

1

Accuracy and Reliability– Manual or spreadsheet-based tracking often causes errors, stock outs, and overstocking, leading to revenue loss.

2

Time & Efficiency– Small business owners waste hours on manual stock checks and reports instead of focusing on customers and growth.

3

Smart Decisions – Lack of analytics prevents businesses from understanding sales trends and demand patterns for better planning.

PROBLEM STATEMENT

Small businesses such as grocery stores, pharmacies, and retail shops still rely on manual records or basic spreadsheets to manage their inventory. This results in:

- Inaccurate stock levels due to human errors in manual entries.
- Frequent stock outs or overstocking, leading to lost sales or wasted resources.
- Time-consuming and inefficient reporting, as invoices and summaries must be prepared manually.

There is a clear need for a simple, automated, and affordable inventory management system that reduces errors, saves time, and empowers small businesses with data-driven decision-making.

FEASIBILITY

The platform simplifies and automates inventory management, enabling small business staff to easily track and update stock in real-time, reducing manual workload and the risk of errors

User-friendly interfaces and barcode scanning features make adoption straightforward, while integrated reporting and alert systems help businesses make faster, data-driven decisions, enhancing day-to-day operations

Integrated analytics and sales reports streamline business decision-making, making inventory management proactive rather than reactive

1

2

3

SOLUTION

To overcome the inefficiencies of manual inventory management, we propose a lightweight, automated Inventory Management System designed specifically for small businesses. The system provides:

- Reorder Alerts to prevent stock outs and overstocking.
- Reporting & Analytics for data-driven decision-making.
- Barcode Integration to reduce manual errors and speed up operations.

This affordable and user-friendly solution empowers small businesses to save time, reduce losses, and operate with enterprise-level efficiency without the complexity of large systems.

Inventory Status Report

Total Products: 5

Low Stock Items: 2

Out of Stock: 0

Product	SKU	Category	Current Stock	Min Threshold	Status	Value
Amoxicillin 500mg	AMX500	Pharmacy	45	20	Good	\$562.50
Rice 5kg Bag	RICE5KG	Grocery	12	15	Low Stock	\$107.88
iPhone Cable	IPHCBL	Electronics	8	10	Low Stock	\$159.92
Bread Loaf	BREAD001	Grocery	25	10	Good	\$74.75
Vitamin D3	VITD3	Pharmacy	30	25	Medium	\$472.50

Stock Alerts

Low Stock Alert

Rice 5kg Bag (RICE5KG) - Current: 12, Minimum: 15

+Restock

Low Stock Alert

iPhone Cable (IPHCBL) - Current: 8, Minimum: 10

+Restock

Barcode Scanner

Click "Start Scanner" to begin scanning

Start Scanner

Or Enter Barcode Manually

Enter barcode number

Search

Top Selling Products

Product	Sales
Bread Loaf	3.0
Amoxicillin 500mg	2.0
iPhone Cable	1.0

DESIGN

A top-down view of a desk with various items: a laptop, a cup of coffee, a pen, a notebook, glasses, and a plant. The text "THANK YOU" is centered in the middle of the image.

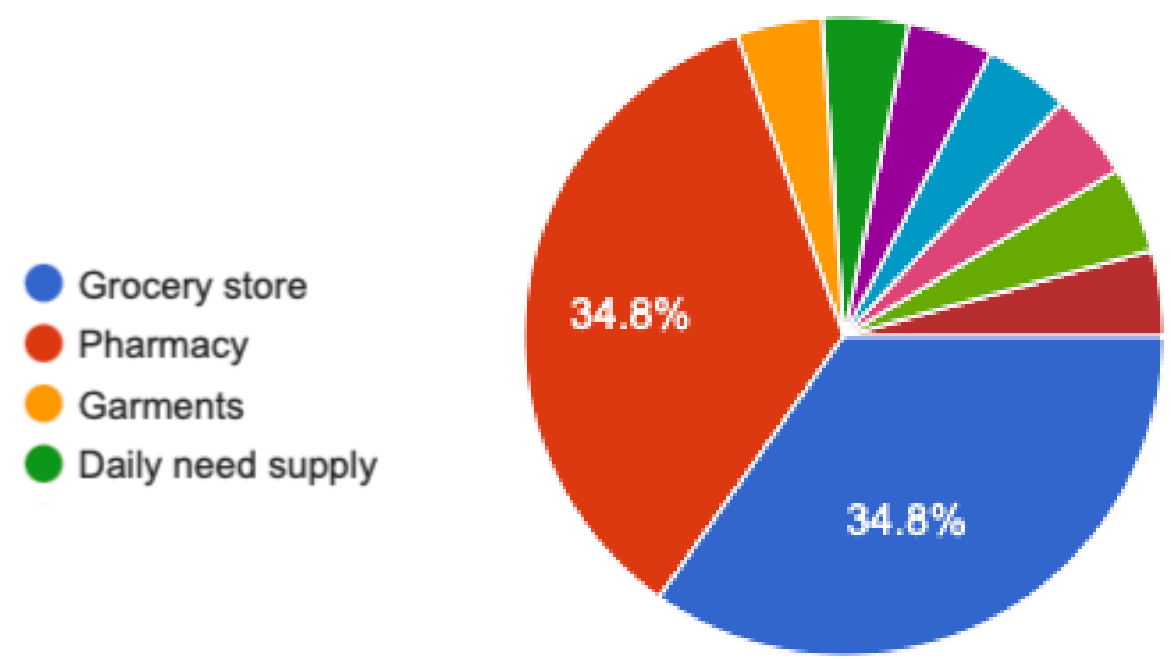
THANK YOU

SMART INVENTORY MANAGEMENT SYSTEM

TEAM - CODE NEXUS

What type of business do you run?

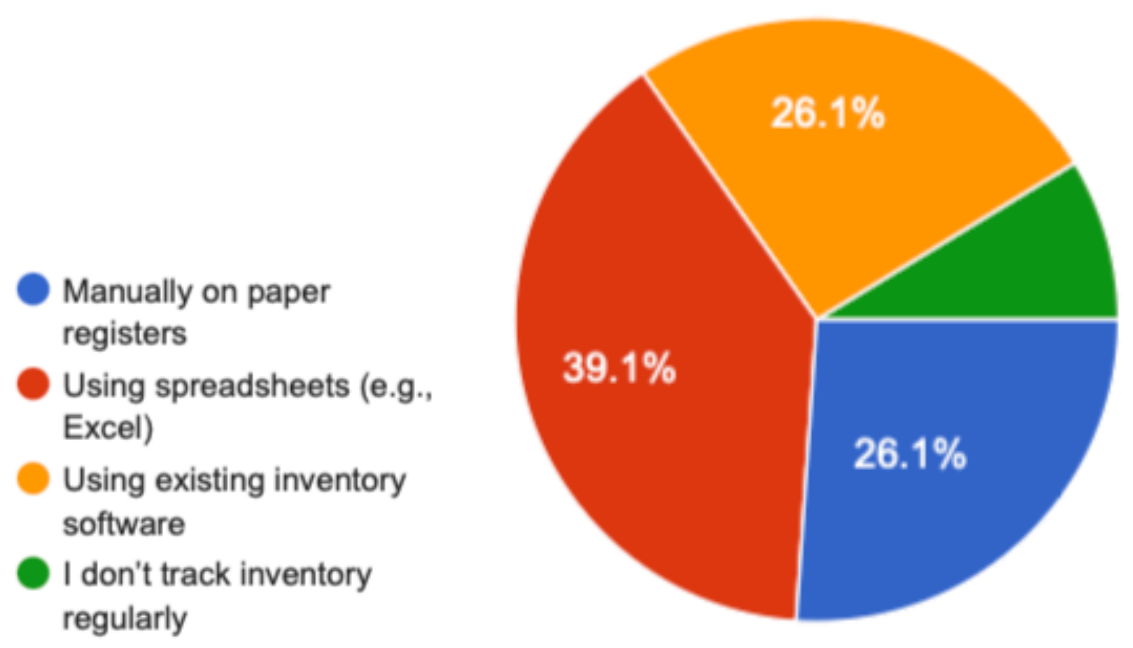
23 responses



- OUR SURVEY COVERS A WIDE SPECTRUM OF SMALL RETAILERS, WITH GROCERY STORES AND PHARMACIES MAKING UP NEARLY 70% OF RESPONDENTS. THIS VALIDATES OUR INITIAL FOCUS ON SECTORS WHERE INVENTORY CHALLENGES ARE MOST URGENT
- NEARLY 40% STILL USE SPREADSHEETS, WHILE OVER A QUARTER RELY ON MANUAL PAPER REGISTERS. ONLY ONE IN FOUR HAS ADOPTED INVENTORY SOFTWARE. A MINORITY, NEARLY 9%, DO NOT TRACK INVENTORY AT ALL.
- THIS INDICATES A CLEAR GAP IN DIGITAL ADOPTION AND HIGHLIGHTS SIGNIFICANT PAIN POINTS IN CURRENT MANUAL/SEMI-MANUAL SYSTEMS—INEFFICIENCY, ERRORS, AND WASTED TIME.
- ONLY 13% MANAGE FEWER THAN 100 ITEMS. THIS SHOWS THAT EVEN SMALL SHOPS NEED SYSTEMS CAPABLE OF SCALING TO HUNDREDS OR THOUSANDS OF PRODUCTS.

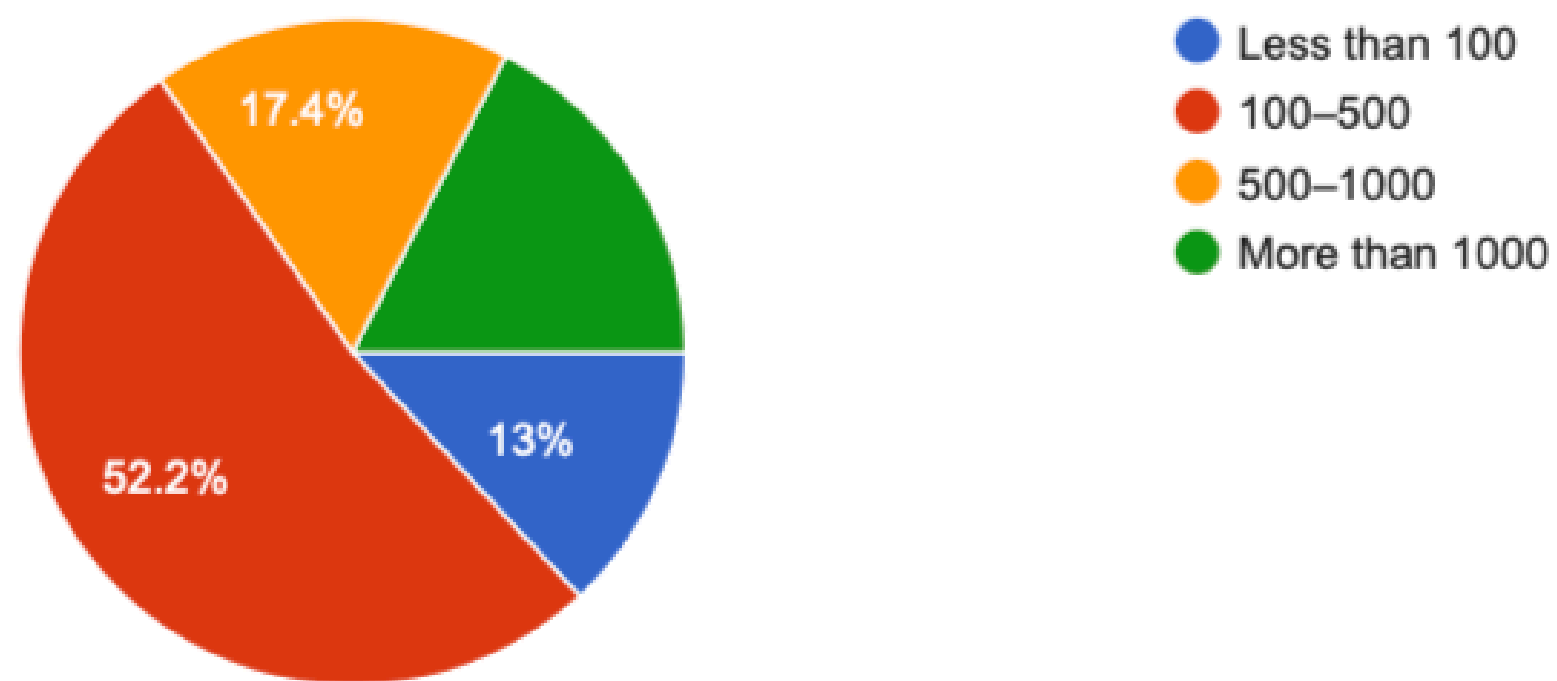
How do you currently manage your inventory?

23 responses



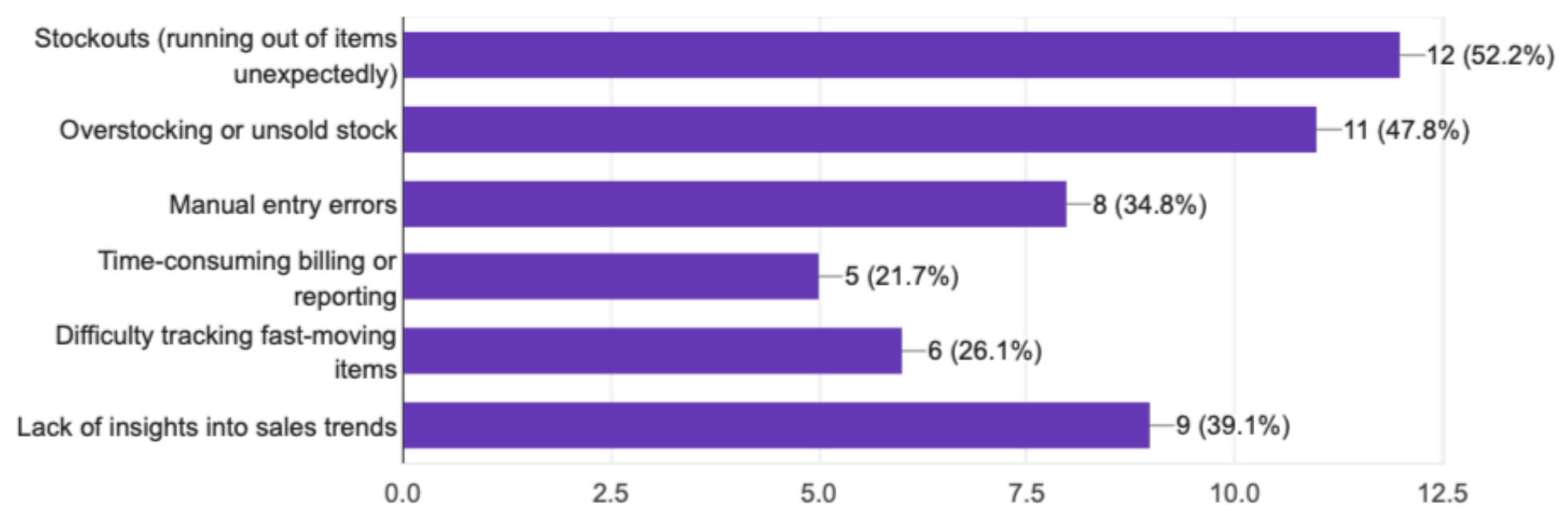
How many products/items do you manage on average?

23 responses



Which of the following inventory challenges do you face most often?
(Select all that apply)

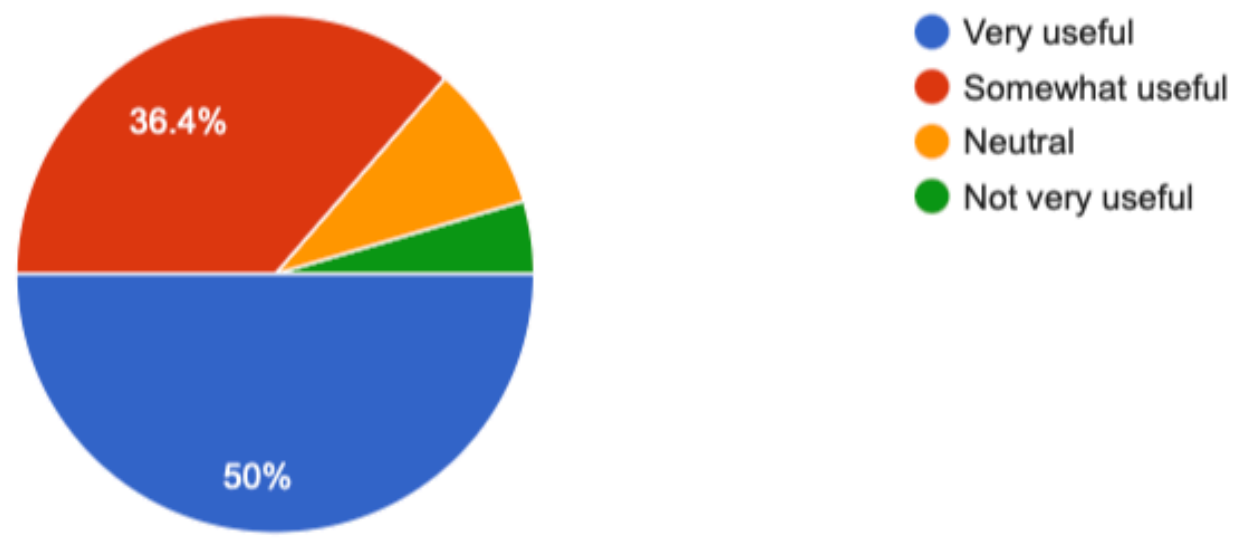
23 responses



- STOCKOUTS AND OVERSTOCKING ARE THE BIGGEST CHALLENGES FOLLOWED BY MANUAL ENTRY ERRORS AND LACK OF INSIGHTS INTO SALES TRENDS AFFECT AROUND 39% WHILE 22% CITE TIME-CONSUMING REPORTING.
- MAJORITY SEE VALUE IN AUTOMATIC REORDER ALERT FEATURE USEFULNESS.

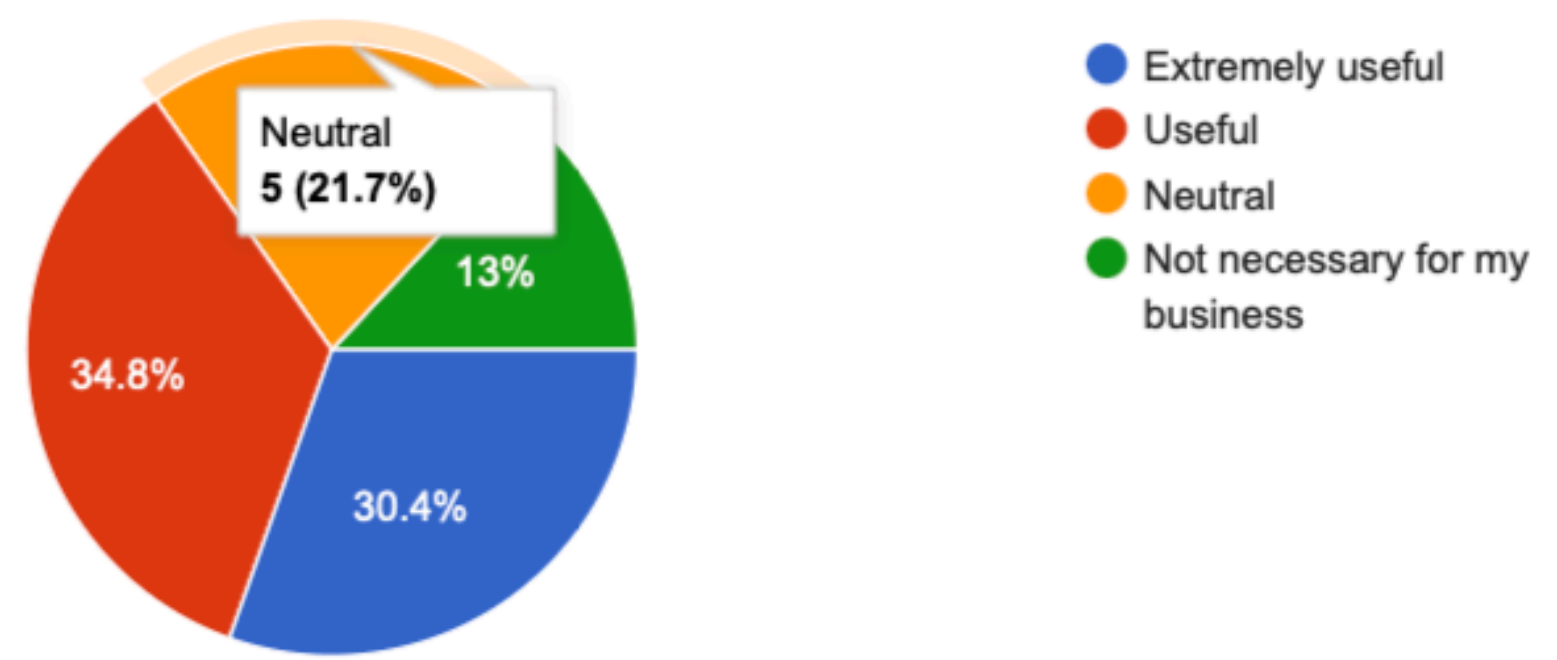
How useful do you think the **Reorder Alert** feature (alerts when stock is low) would be for your business?

22 responses



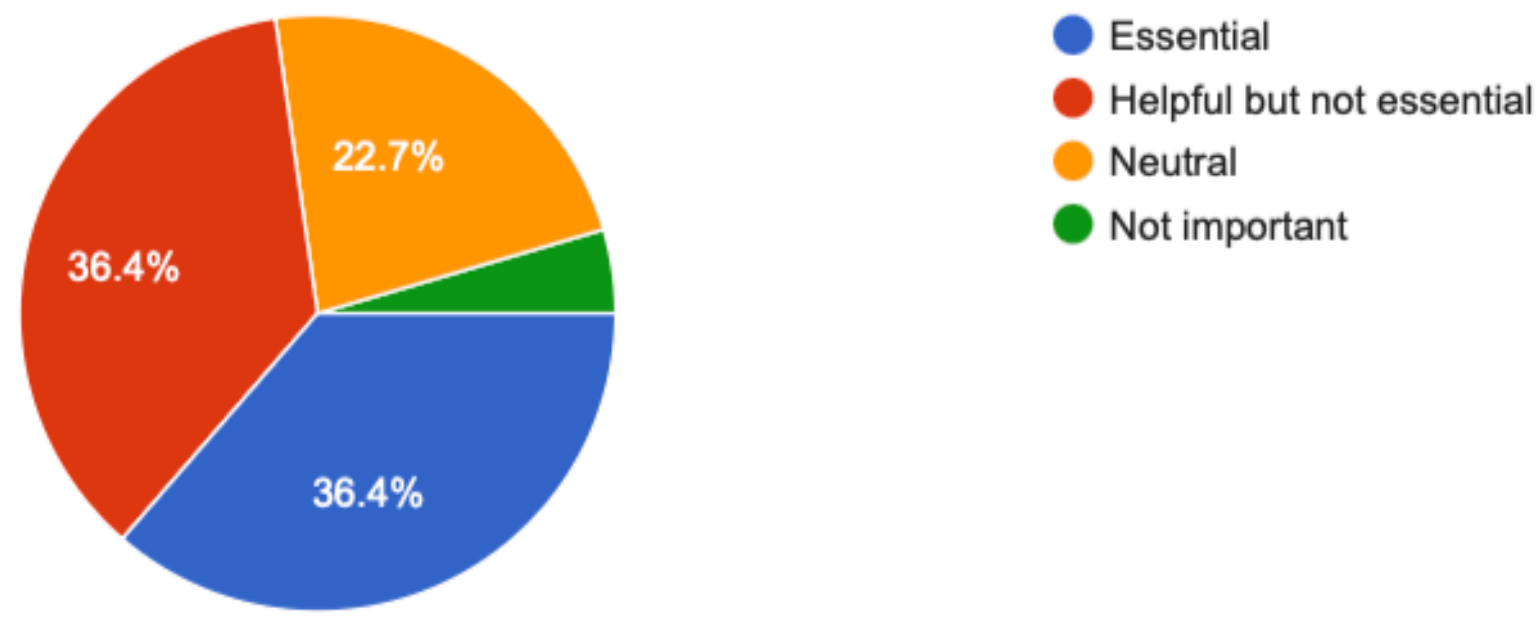
How useful is **Barcode Scanning Integration** for faster checkouts and inventory updates?

23 responses



How important are **automated reports and analytics** in your daily business decision-making?

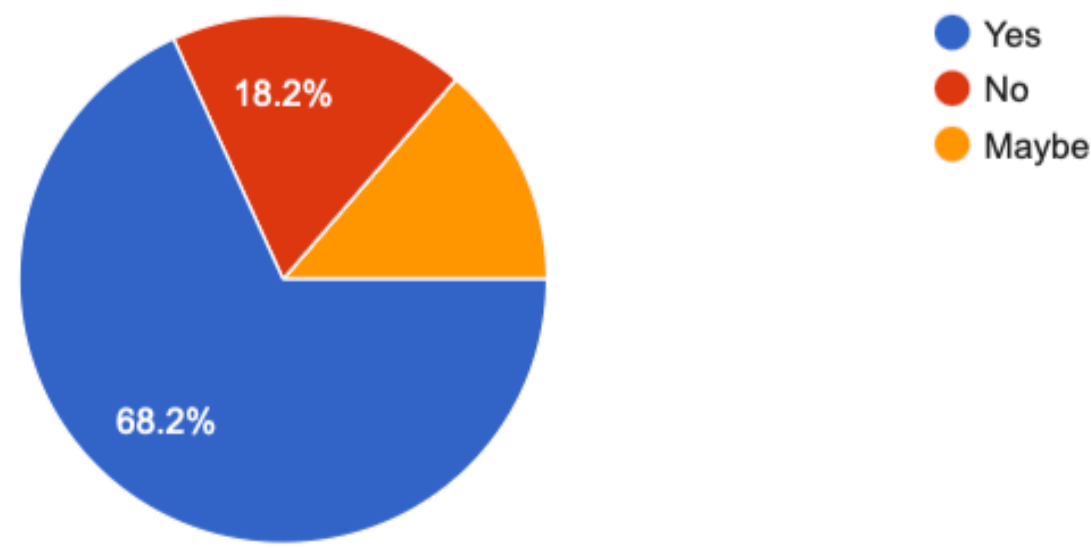
22 responses



- A STRONG MAJORITY (65%) OF SHOPKEEPERS WANT INTUITIVE DASHBOARDS DISPLAYING TOP-SELLING PRODUCTS, STOCK STATUS, AND SALES TRENDS. VERY FEW FIND THEM UNHELPFUL, CONFIRMING BROAD APPEAL ACROSS YOUR USER BASE.

Would you find **Dashboard Analytics** (showing top-selling products, stock status, and trends) valuable?

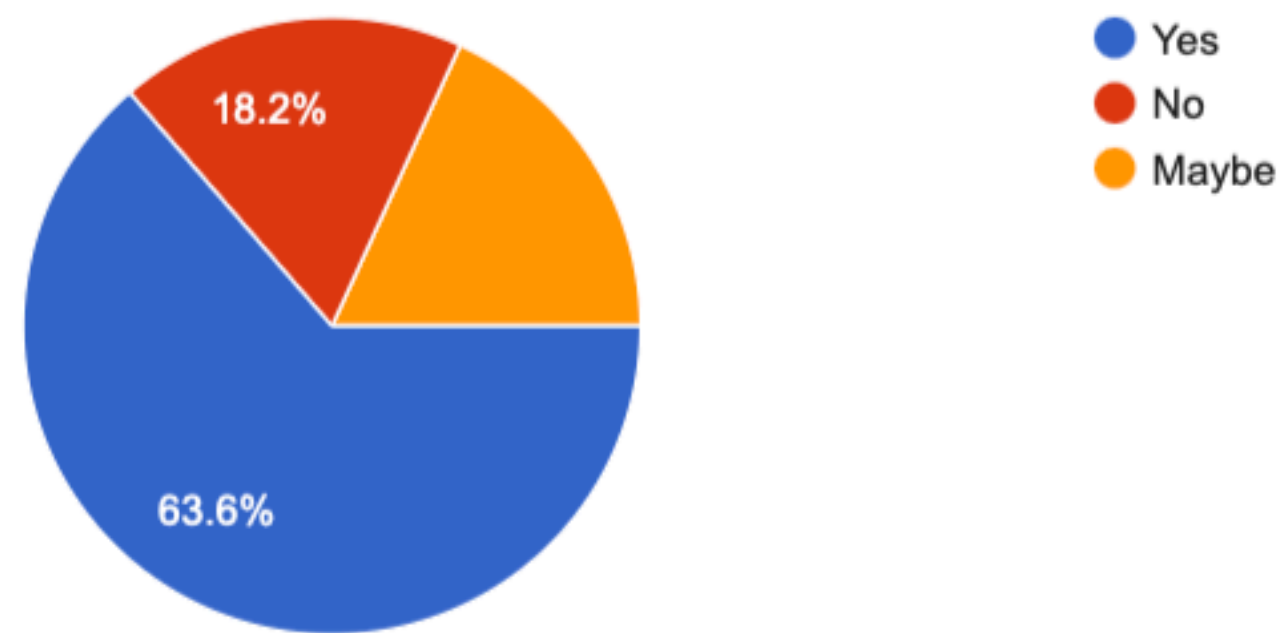
22 responses



- SHOPKEEPERS LACK TIME FOR COMPLEX DATA ANALYSIS—VISUAL, EASY-TO-UNDERSTAND DASHBOARDS HELP THEM MAKE QUICK, SMARTER DECISIONS ABOUT REORDERING, PROMOTIONS, AND WHAT PRODUCTS TO DISCONTINUE.

Do you think implementing this system would help reduce manual work and save time?

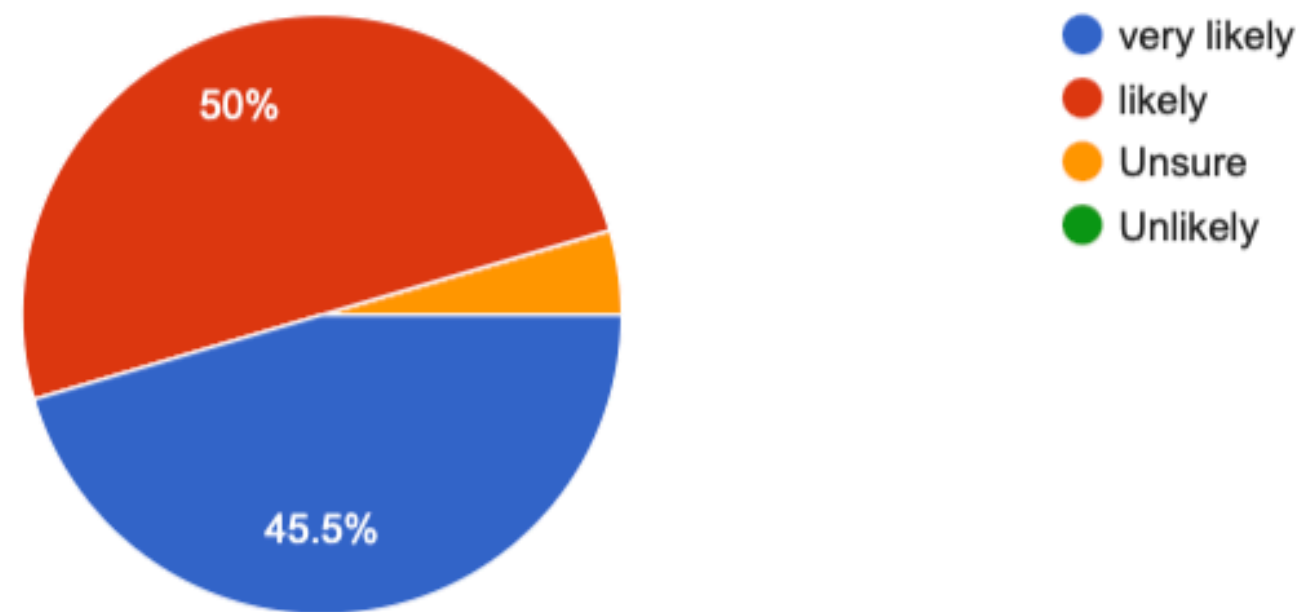
22 responses



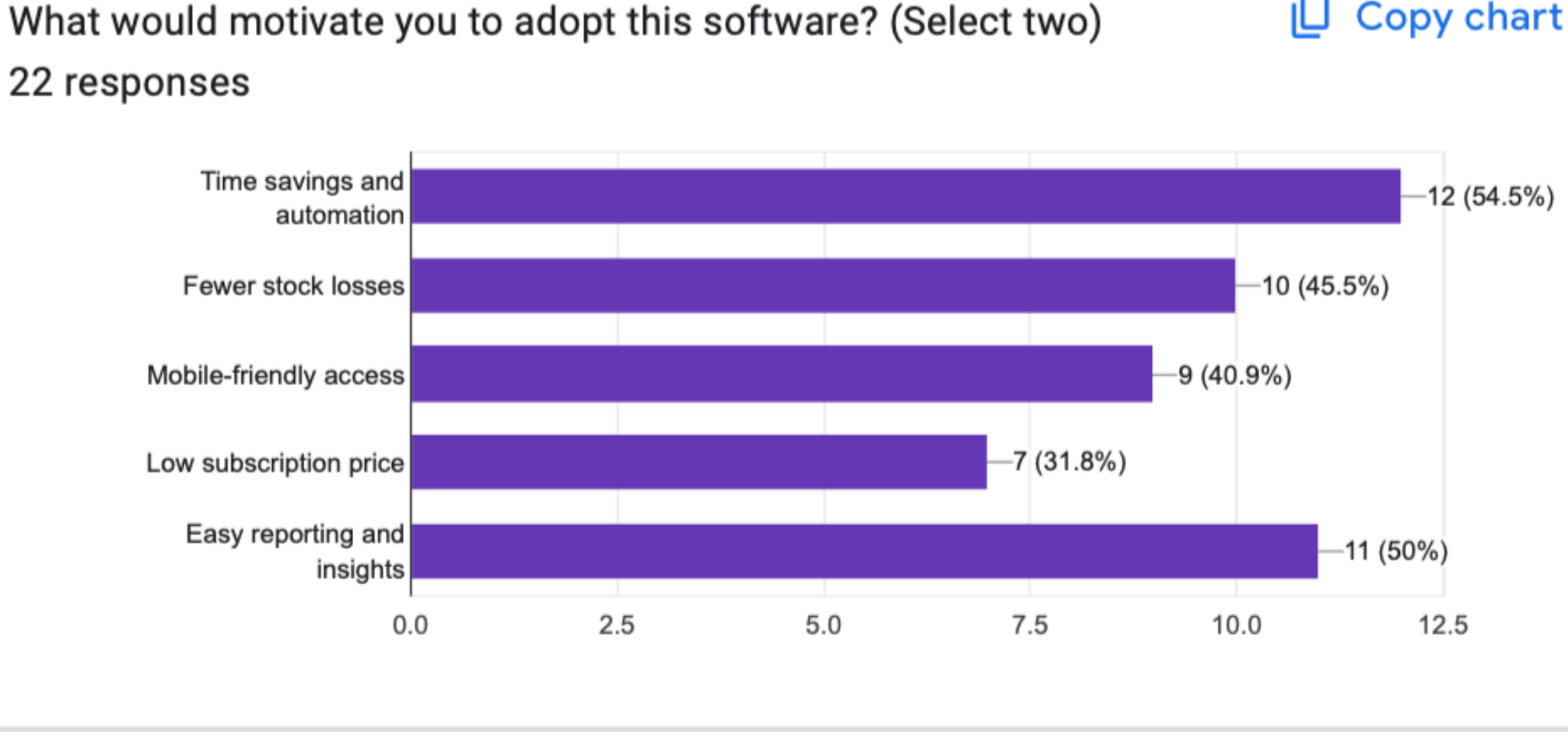
- A LARGE MAJORITY (65%) OF SHOPKEEPERS AGREE THAT IMPLEMENTING THIS SYSTEM WOULD HELP REDUCE MANUAL WORK AND SAVE TIME, VALIDATING THAT AUTOMATION ADDRESSES A REAL PAIN POINT FOR SMALL BUSINESSES.

How likely are you to use such a system if it is affordable and easy to use?

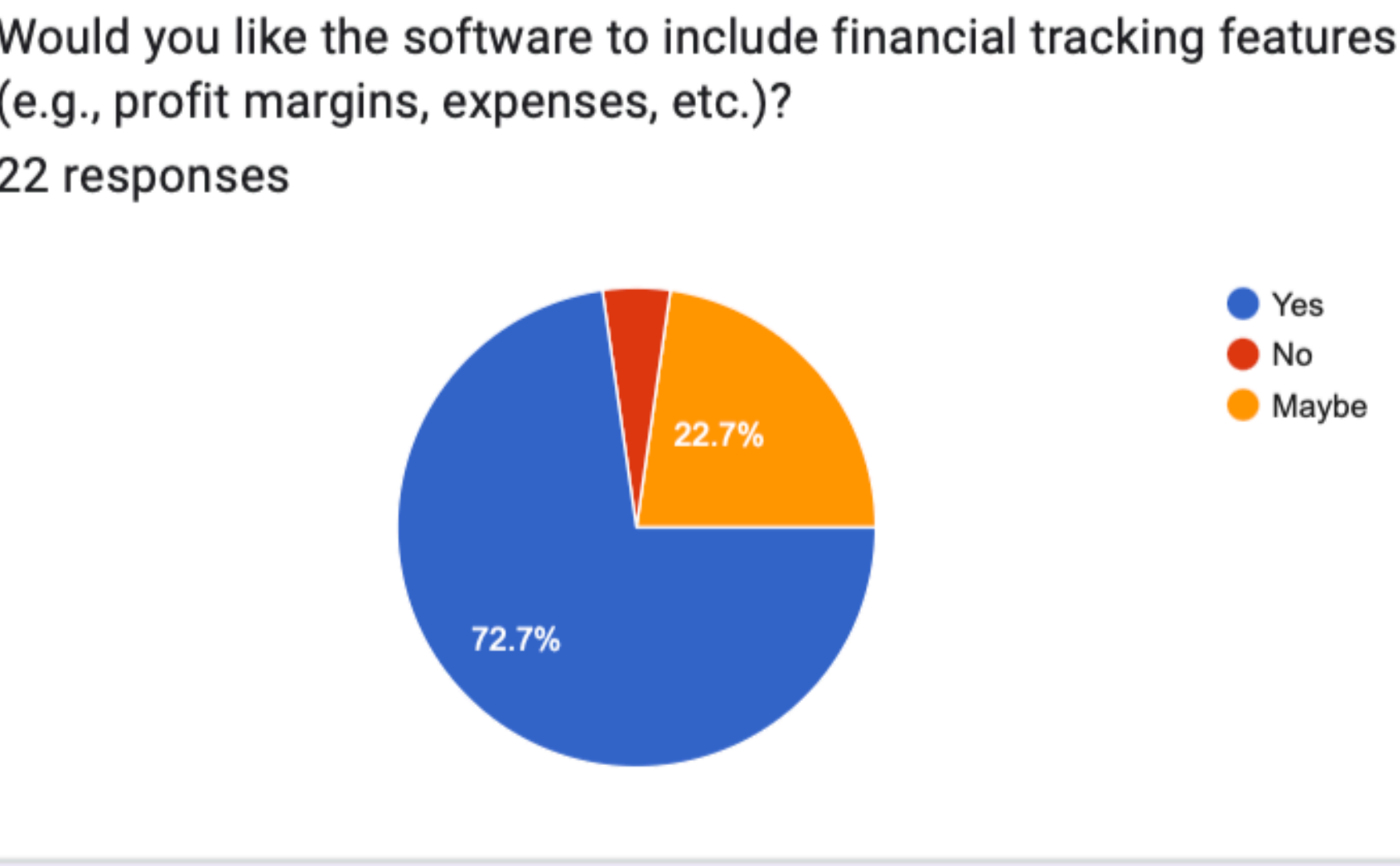
22 responses



- OVER 90% ARE AT LEAST LIKELY OR VERY LIKELY TO USE THE SYSTEM—IF IT IS AFFORDABLE AND EASY TO USE—SHOWING STRONG MARKET INTEREST PROVIDED COSTS AND USABILITY BARRIERS ARE ADDRESSED.



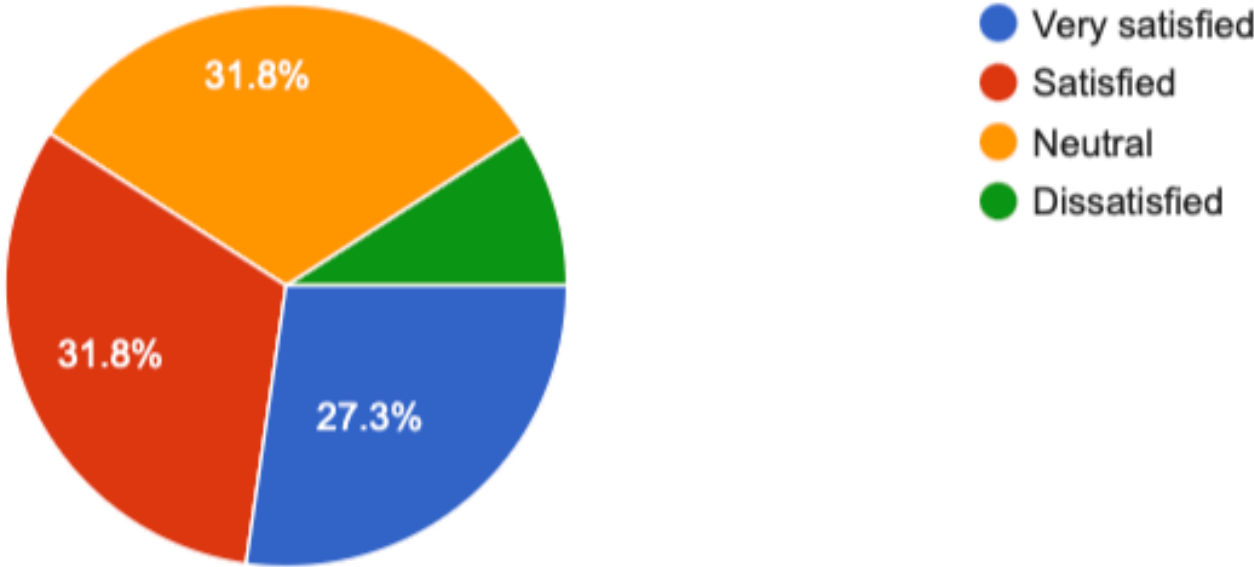
- THE TOP MOTIVATORS FOR ADOPTING YOUR SOFTWARE ARE TIME SAVINGS AND AUTOMATION, AS WELL AS EASY REPORTING AND ACTIONABLE INSIGHTS (BOTH CHOSEN BY 52% OF USERS).



- OTHER STRONG FACTORS INCLUDE FEWER STOCK LOSSES AND MOBILE-FRIENDLY ACCESS (EACH 43.5%), HIGHLIGHTING THE NEED FOR CONVENIENCE AND LOSS PREVENTION FOR SMALL SHOPKEEPERS.

How satisfied are you with your current inventory management process?

22 responses



What improvements or features would you like to see in an inventory management system?

5 responses

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- Real-Time Predictive Forecasting: Use AI to predict demand, not just track current stock. (No more "Whoops, sold out.")
- Seamless Multi-Channel Sync: Effortlessly integrate with all sales platforms (e-commerce, marketplaces, POS).
- Mobile-First Barcode Management: Allow staff to handle all core tasks (receiving, picking) using a smartphone camera, even offline.
- More user friendly
- easy inventory tracking

If given a free trial for this system, would you be interested in testing it?

22 responses

